

DEPARTMENT OF INFORMATION TECHNOLOGY

Machine Learning LAB
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Diabetes Prediction Dashboard

GUI-Based Machine Learning Platform for Healthcare Classification

Group Number: 12



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Chapter 1

Introduction

Machine Learning is one of the most important technologies used for prediction and data analysis in modern applications. In healthcare, Machine Learning helps in identifying diseases using patient medical data and predictive models.

This project presents a GUI-based Machine Learning application named **Diabetes Prediction Dashboard**. The application is developed using Streamlit and Scikit-learn and focuses on diabetes prediction using classification algorithms.

The system allows users to train, evaluate, compare, and visualize different Machine Learning algorithms through an interactive graphical user interface. The dashboard simplifies Machine Learning analysis and provides meaningful visualizations for understanding dataset behavior and model performance.

Chapter 2

Objectives

The main objectives of this project are:

- To develop a graphical user interface for Machine Learning algorithms
- To implement diabetes prediction using classification models
- To visualize dataset analysis using graphs
- To compare multiple Machine Learning algorithms
- To implement hyperparameter tuning
- To evaluate model performance using different metrics

Chapter 3

Technologies Used

Component	Technology Used
Programming Language	Python
GUI Framework	Streamlit
Machine Learning Library	Scikit-learn
Visualization Libraries	Matplotlib, Seaborn, Plotly
Data Processing	Pandas, NumPy

Chapter 4

Dataset

The project uses the **PIMA Indians Diabetes Dataset** for binary classification. The dataset contains medical information of patients and is used to predict whether a patient is diabetic or non-diabetic.

4.1 Dataset Features

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Blood pressure value
SkinThickness	Skin fold thickness
Insulin	Insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes hereditary score
Age	Age of patient
Outcome	0 = No Diabetes, 1 = Diabetes

Chapter 5

GUI Features

The project contains an interactive graphical user interface developed using Streamlit.

5.1 Dataset Section

The dataset section provides:

- Dataset preview
- Statistical summary
- Missing value analysis
- Target distribution

5.2 Visualization Section

The visualization section contains graphical analysis of the dataset.

Included visualizations:

- Correlation Heatmap
- Histogram
- Scatter Plot
- Box Plot
- Violin Plot

5.3 Training Section

The training section allows users to:

- Select Machine Learning algorithms
- Tune hyperparameters
- Train models interactively

5.4 Results Section

The results section displays:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve
- Classification Report

5.5 Compare Models Section

This section compares different Machine Learning models using cross-validation accuracy.

Chapter 6

System Workflow

The project follows the following workflow:

1. Load dataset
2. Preprocess dataset
3. Scale features
4. Split dataset into training and testing sets
5. Select Machine Learning algorithm
6. Tune hyperparameters
7. Train model
8. Evaluate model performance
9. Visualize results
10. Compare algorithms

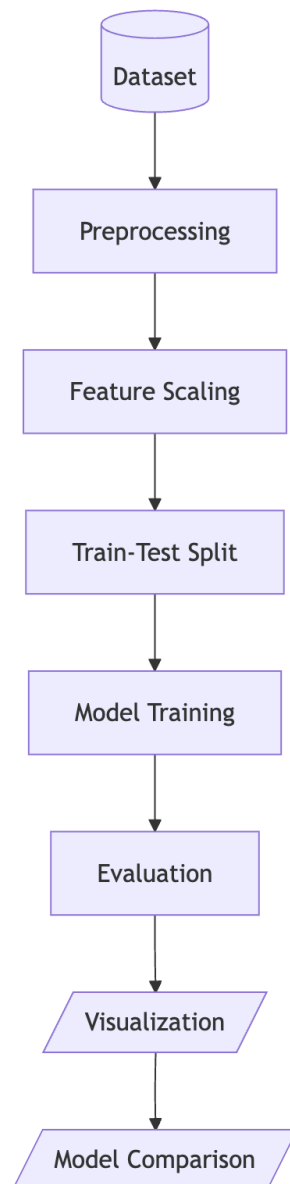


Figure 6.1: System Workflow

Chapter 7

Machine Learning Algorithms

7.1 Logistic Regression

Logistic Regression is a supervised Machine Learning algorithm used for binary classification problems. It predicts the probability of a class using the sigmoid function.

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

7.2 Decision Tree

Decision Tree is a tree-based classification algorithm that splits data into branches based on feature conditions.

7.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting.

7.4 K-Nearest Neighbors (KNN)

KNN classifies data points based on the nearest neighboring samples.

7.5 Support Vector Machine (SVM)

SVM separates classes using hyperplanes and maximizes the margin between classes.

7.6 Naive Bayes

Naive Bayes is a probability-based Machine Learning algorithm based on Bayes theorem.

Chapter 8

Data Preprocessing

Data preprocessing is an important step in Machine Learning.

The following preprocessing techniques are used in the project:

8.1 Missing Value Analysis

The dataset is checked for missing values before training the models.

8.2 Feature Scaling

Feature scaling is performed using `StandardScaler` to normalize numerical values and improve model performance.

8.3 Train-Test Split

The dataset is divided into training and testing sets for model evaluation.

Chapter 9

Hyperparameter Tuning

The dashboard allows users to adjust various Machine Learning parameters interactively.

- K value for KNN
- Max Depth for Decision Tree
- Number of Estimators for Random Forest
- Kernel type for SVM
- C value for Logistic Regression and SVM
- Test size percentage

Hyperparameter tuning improves model performance and allows users to experiment with different configurations.

Chapter 10

Evaluation Metrics

The following evaluation metrics are used in the project.

10.1 Accuracy

Accuracy represents the percentage of correctly predicted samples.

10.2 Precision

Precision measures how many predicted positive samples are actually positive.

10.3 Recall

Recall measures the ability of the model to correctly identify positive samples.

10.4 F1 Score

F1 Score is the harmonic mean of precision and recall.

10.5 Confusion Matrix

Confusion Matrix displays:

- True Positive
- True Negative
- False Positive
- False Negative

10.6 ROC Curve

ROC Curve is used to evaluate classification performance at different threshold values.

Chapter 11

Visualizations

The project contains multiple graphical representations for data analysis and model evaluation.

11.1 Correlation Heatmap

Heatmap shows the relationship between different dataset features.

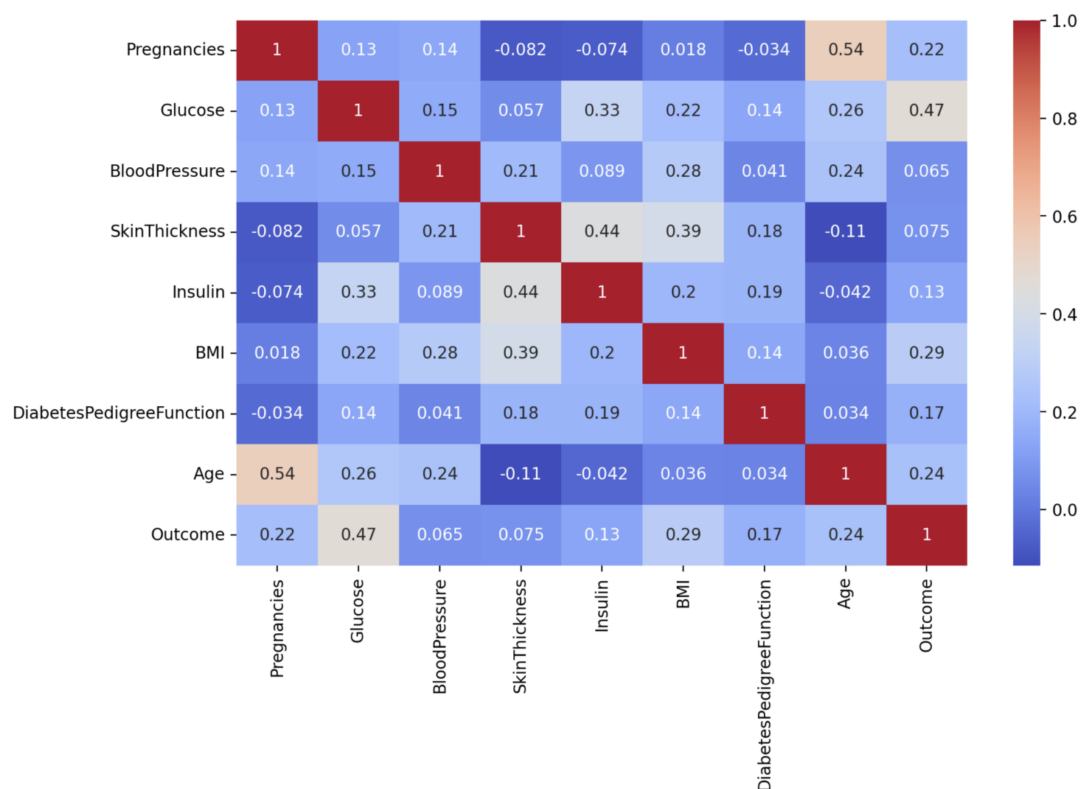


Figure 11.1: Correlation Heatmap

11.2 Histogram

Histogram shows feature distribution.

11.3 Scatter Plot

Scatter plots help in understanding relationships between features.

11.4 Box Plot

Box plots help identify outliers in the dataset.

11.5 Confusion Matrix

Confusion Matrix visualizes prediction performance.

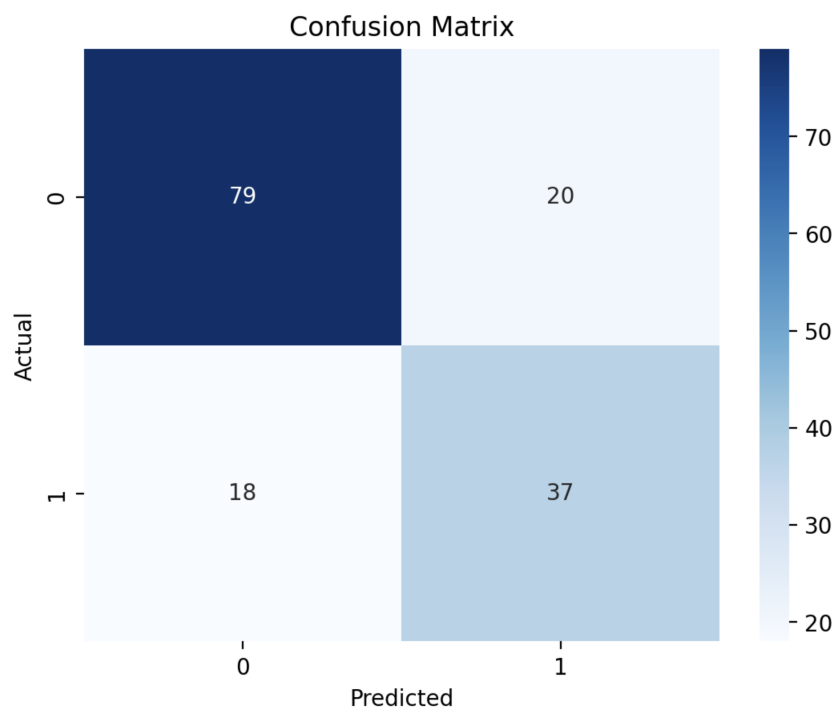


Figure 11.2: Confusion Matrix

11.6 ROC Curve

ROC Curve evaluates classification quality.

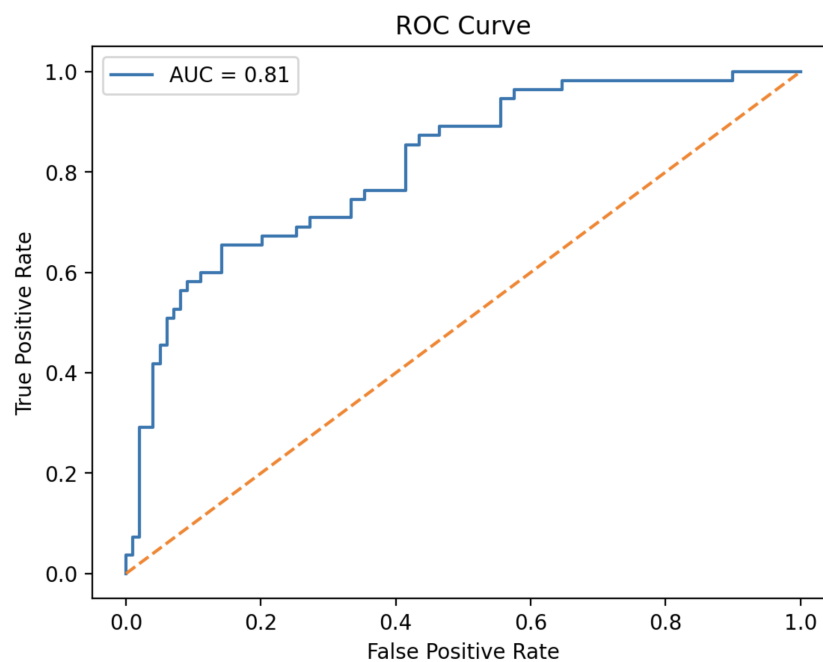


Figure 11.3: ROC Curve

11.7 Model Comparison Graph

The graph compares cross-validation accuracy of different Machine Learning algorithms.

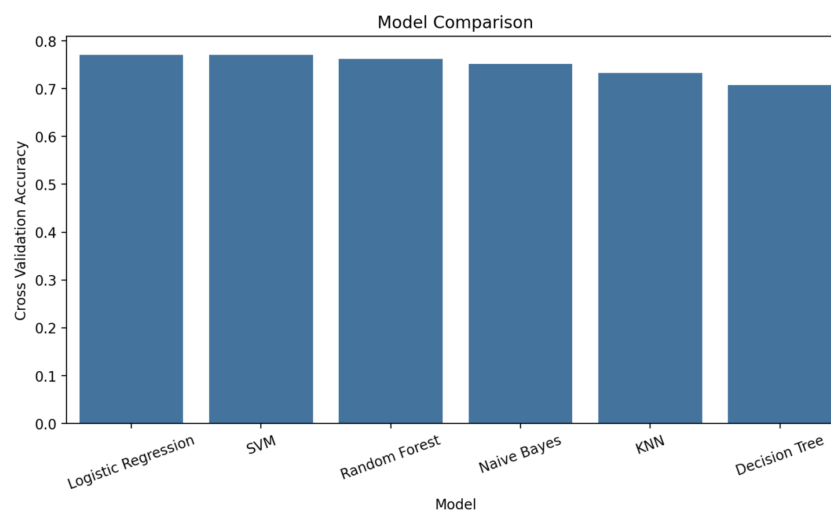


Figure 11.4: Model Comparison

Chapter 12

Results and Discussion

Different Machine Learning algorithms were trained and evaluated using the diabetes dataset.

The models successfully classified diabetic and non-diabetic patients using healthcare features.

Among different algorithms, Random Forest and SVM provided better classification performance compared to other models.

The graphical visualizations helped in understanding feature relationships and model performance effectively.

12.1 Performance Metrics

Metric	Value
Accuracy	84%
Precision	82%
Recall	80%
F1 Score	81%

12.2 Model Comparison

Model	Accuracy
Logistic Regression	78%
Decision Tree	76%
Random Forest	84%
KNN	80%
SVM	82%

Naive Bayes	75%
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Chapter 13

Advantages

- User-friendly graphical interface
- Interactive visualizations
- Multiple Machine Learning algorithms
- Hyperparameter tuning support
- Real-time model comparison
- Easy understanding of Machine Learning concepts

Chapter 14

Future Scope

The project can be improved further by adding:

- Multi-disease prediction
- Deep learning models
- Cloud deployment
- Real-time healthcare analytics
- Mobile application support

Chapter 15

Conclusion

The Diabetes Prediction Dashboard successfully demonstrates the implementation of Machine Learning classification algorithms using an interactive graphical user interface.

The project integrates data preprocessing, visualization, model training, hyperparameter tuning, model comparison, and performance evaluation within a single platform.

The developed application satisfies all major requirements of a Machine Learning laboratory project and provides an efficient way to understand Machine Learning concepts through healthcare data analysis.

References

1. Scikit-learn Documentation
2. Streamlit Documentation
3. Python Documentation
4. PIMA Indians Diabetes Dataset
5. Matplotlib Documentation
6. Seaborn Documentation